

For the U-M ECT consultation

Dad's clinical context + questions for the team · revised 2026-05-20

For Dad, Mom, Leah, David,
Dr. Rubin, Dr. Chilton, Julane,
family team

What this document is for. A one-page brief that gives Dad and Mom the clinical context to share with the U-M ECT team and the questions worth discussing. **The handout is not arguing against ECT.** The published evidence on ECT in patients like Dad is actually reasonably favorable (Morikawa 2024 in confirmed Lewy-spectrum disease: 96% improvement, no cognitive worsening). The handout's job is to make sure the U-M team has the relevant context and that the family is informed about the genuine tradeoffs — so the right decision can be made by Dad and the clinical team together, not pushed in either direction by the family.

DAD'S CLINICAL CONTEXT TO SHARE WITH THE U-M ECT TEAM

- **Severe TRD (PHQ-9 = 24), worst episode in 40 years.** Currently on duloxetine 60 mg, doxepin 6 mg HS, trazodone 50 mg PRN, melatonin 3 mg HS. Activation-intolerance pattern across six prior 2025 psychotropics (aripiprazole, mirtazapine, brexpiprazole, olanzapine, pramipexole, lithium). Lorazepam taper recently completed.
- **February 2026 IV ketamine course produced a dramatic ~4-week response** on duloxetine + brexpiprazole. Dad is a documented IV-ketamine responder. A2 ketamine clinic has him scheduled to restart June 8 — though the May 18 first session at Royal Oak was difficult and Dad's appetite for another ketamine session is uncertain.
- **March 2026 Aquablation BPH procedure under general anesthesia** (3/18) was followed by depression return ~1–2 weeks later, new motor and expression changes by 4/8, and gait abnormality + decreased appetite + unintentional weight loss coded at the 4/14 PCP visit. The temporal correlation is suggestive but not definitively post-operative delirium or POCD; depression relapse and ketamine cessation overlap the same window.
- **Prodromal-synucleinopathy markers (some predating the March GA).** Formal REM Sleep Behavior Disorder (G47.52) since 10/2017 (8.5 years). Balance problem coded 7/2025 — 9 months before the March procedure. Chronic constipation. Peripheral neuropathy on chart since 2018. Appetite-interoception dissociation for 1–2 years.
- **U-M Movement Disorders evaluation pending**, including DAT-SPECT. Results would meaningfully inform ECT technique and monitoring — Morikawa 2024's favorable ECT-in-Lewy-spectrum data came from patients with confirmed abnormal DAT-SPECT.
- **Cardiac:** Left anterior fascicular block + sinus bradycardia (HR 49–53), BP 112/58, QTcF 401 (5/19/2025 EKG). Relevant for ECT-induced bradycardia/asystole risk.

QUESTIONS WORTH DISCUSSING WITH THE U-M TEAM

- **What is the team's reading of ECT-vs-another-ketamine-course for Dad specifically**, given his February response, the difficulty of the May 18 session, and his current preferences? Both are reasonable choices; the family wants to understand the team's perspective on the relative merits.
- **What technique would the team plan?** Right unilateral ultrabrief pulse minimizes cognitive side effects and bradycardia risk per Stewart 2011. Propofol-based anesthesia (vs methohexital) has a lower inflammatory profile and faster cognitive recovery in elderly per Pang 2021. Worth confirming the plan upfront.
- **Should ECT wait for the pending DAT-SPECT result?** If abnormal, that confirms Dad in the Morikawa 2024 patient population with documented favorable ECT outcomes, and would inform technique. If normal, the

prodromal-synucleinopathy concern attenuates. The result is likely available within 1–3 weeks.

- **What does the team see as the realistic response and durability expectations** in Dad's specific profile? Late-life severe TRD response rates with ECT are 60–70% per Kellner PRIDE; Morikawa 2024 showed 96% improvement in Lewy-spectrum patients but 23.8% recurrence at 12 months. What does the team expect for Dad, and what's the maintenance plan if he responds?
- **What's the plan for cognitive baseline and serial monitoring?** Pre-ECT MoCA and Trail Making B baseline, with serial monitoring every 3–4 sessions, is the standard of care. Dad has documented brain fog and pending cognitive evaluation — establishing baseline before any ECT-related decline is critical.
- **How does the team think about the March anesthesia history?** The temporal correlation between Dad's Aquablation GA and the subsequent decline raised concerns about a possible vulnerability signal. The team's read on whether this should affect ECT technique, monitoring intensity, or timing would be useful.

A QUESTION DAD OR MOM MIGHT PUT TO THE TEAM TO OPEN THE CONVERSATION

FOR DAD OR MOM — IN THEIR OWN WORDS

Dr. R and Dr. Auchincloss have both recommended ECT, and we're trying to understand whether it's the right next step for Dad specifically. We know the literature on ECT in Lewy-spectrum patients is actually pretty favorable (Morikawa 2024). The questions we have are about timing relative to the pending DAT-SPECT and Movement Disorders workup, about technique, and about how the team thinks about the alternative of completing the IV ketamine course at A2 first. What's your recommendation?

The honest balance on ECT for Dad's profile

A revised reading of the published evidence (per OE consultation #19, 5/20). The earlier draft of this handout overstated the case against ECT; this revision is more balanced. ECT is a reasonable option for Dad, with both meaningful upsides and real but manageable downsides.

WHAT FAVORS ECT IN DAD'S PROFILE

- **Robust late-life severe TRD evidence:** Kellner PRIDE 2016 (n=240, age ≥ 60 , right unilateral ultrabrief pulse + venlafaxine) showed **61.7% remission** at end of acute course. Late-life ECT response rates: 60–70%.
- **Specifically in Lewy-spectrum disease:** Morikawa 2024 (n=25 confirmed DLB / prodromal DLB with abnormal DAT-SPECT) showed **96% at least some improvement, 60% remarkable improvement**. Parkinsonism actually improved with ECT. No cognitive worsening. This is the most directly applicable safety data for Dad.
- **Takamiya 2021** meta-analysis of ECT in Parkinson's disease (14 studies, n=129): improved both motor and non-motor symptoms without cognitive worsening. No documented cases of ECT precipitating motor decline in prodromal-synucleinopathy.
- **Fast onset:** response typically detectable within 1–2 weeks. Faster than starting any new oral antidepressant.
- **Cognitive cost is largely transient in responders.** ELEKT-D showed 8.8-point delayed-recall decrement at end of treatment, but Kumpf 2025 secondary analysis showed differences resolved among responders at 1-, 3-, and 6-month follow-up.
- **If Dad doesn't feel up for another ketamine session,** ECT is a legitimate alternative — not a last resort, just a different treatment with its own evidence base.

WHAT ARGUES FOR CAUTION OR FOR KETAMINE-FIRST

- **ELEKT-D 2023** (n=403): ketamine non-inferior to ECT for nonpsychotic TRD with significant cognitive-safety advantage during the treatment course. If Dad is open to another ketamine attempt, this argues for trying that first.
- **Documented prior ketamine response** (Feb 2026, ~4-week dramatic response). N=1 prior response is the single strongest predictor of re-response. The May 18 Royal Oak session was difficult but is not a fairly-tested ketamine course at A2.
- **GA exposure considerations.** Per-ECT-session GA is brief (~5–10 min TIVA) and lower per-exposure risk than the March 90-min surgical GA. Cumulative across induction (6–12 sessions) is modest. Maintenance ECT (20–40 sessions/year if pursued) is where cumulative-dose concerns become meaningful — though this only arises if induction is successful.
- **The March 2026 post-GA recovery has been imperfect.** Whether the deterioration was POD/POCD or depression relapse + life situational + ketamine cessation is genuinely unclear. Either interpretation suggests some elevated vulnerability — but the magnitude is uncertain.
- **The pending DAT-SPECT** would materially refine the picture. If abnormal, it confirms Dad in the Morikawa 2024 cohort (good ECT outcomes). If normal, the prodromal-synucleinopathy concern attenuates. Worth a few weeks' wait if clinically feasible.
- **Acuity does not currently meet the published threshold** for ECT urgency (suicidality, catatonia, psychosis, food refusal). This doesn't argue against ECT — it argues that the family has time to do the workup and decision-making properly rather than under emergency pressure.

HOW SEVERE ARE THE GA EXPOSURES, REALLY?

A more precise read than the earlier version of this handout offered:

- **Per ECT session:** ~5–10 minutes of total intravenous anesthesia (propofol or methohexital + succinylcholine for muscle relaxation). No volatile inhalation agents, no surgical inflammation, no blood loss, no post-op opioids. **Per-exposure cognitive risk is materially lower than what a 60–90 minute surgical GA (like the March Aquablation) represents.** Not zero — but not in the same magnitude.

- **Cumulative across an induction course (6–12 sessions):** Total GA time ~30–120 minutes across 3–4 weeks. The MAAS cohort's cumulative-time signal applies but at the lower end of the dose-response. The acute cognitive cost during the course (ELEKT-D's 8.8-point delayed-recall decrement) largely **resolves in responders by 1 month**.
- **Cumulative across induction + a year of maintenance:** 20–40 brief GA exposures total. This is the actually-concerning cumulative dose, but it only arises if Dad responds and is committed to maintenance. Maintenance frequency can be adjusted; many patients taper to monthly or less; some don't need maintenance at all. This is a downstream question, not a present-decision question.
- **The Dad-specific risk multiplier:** Kim 2018 found RBD as POD predictor (OR 1.45) in elderly surgical patients. Dad's prodromal-synucleinopathy profile theoretically raises POD risk. But Morikawa 2024 — the population that overlaps Dad's profile most — found 24% transient delirium rate and no serious AEs across the full course. The Dad-specific multiplier is real but not large.

ECT vs another ketamine course — comparing honestly for Dad

This is the actual decision the family is helping Dad weigh. Both are legitimate evidence-supported choices for severe late-life TRD. The right answer depends substantially on Dad's preferences and the specifics of his profile.

Dimension	ECT at U-M	Another IV ketamine course at A2 (June 8)
Response rate in Dad's profile	60–70% in late-life severe TRD per PRIDE; 96% any improvement, 60% remarkable in Lewy-spectrum per Morikawa 2024	~50% in elderly TRD (TRANSFORM-3 esketamine ≥65 was negative; IV ketamine elderly evidence sparser); but Dad's documented prior dramatic response elevates this substantially
Onset of effect	1–2 weeks typical for ECT in TRD	Within days to first week typical for IV ketamine in prior responders
Cognitive cost during treatment	Real but typically transient — 8.8-point delayed-recall decrement (ELEKT-D), resolves by 1 month in responders	Minimal — 0.9-point decrement on same measure (ELEKT-D)
GA exposure	6–12 brief TIVA exposures over the induction course; 20–40 over a year if maintenance is needed	None — subanesthetic dosing, no GA
Patient experience	Brief sedation, generally not remembered. Less of the dissociative experience Dad found difficult on 5/18.	Dissociative experience during the infusion; the 5/18 Royal Oak session was hard. Could be different at A2 with Julane present, but cannot be guaranteed.
Durability	Maintenance often required; 23.8% recurrence at 12 months in Lewy-spectrum (Morikawa); 61.5% psychiatric readmission at 2 years (Kobayashi 2026 LBD)	Maintenance also required; ~46% relapse without scheduled maintenance (Phillips 2019); ~27% with structured maintenance
Patient willingness	Dad has not actively resisted ECT; both Rubin and Auchincloss recommend it	Dad found the 5/18 session difficult; current willingness to repeat is uncertain
Setup and logistics	U-M ECT program; established workflow; standard scheduling	A2 clinic with Julane; June 8 scheduled; new setting Dad hasn't yet been to

A DECISION FRAME THAT MAY HELP

- **If Dad actively wants to try ECT** (or feels more able to face it than another ketamine session): the evidence supports proceeding, with attention to technique (right unilateral ultrabrief pulse, propofol-based anesthesia, cognitive baseline and monitoring). The Morikawa 2024 data is reassuring for his profile.
- **If Dad is open to either path:** the published evidence and his prior responder status modestly favor completing the A2 ketamine course first, with ECT held as the next step if the ketamine course doesn't produce a sustained response. The reasoning is the cognitive-safety advantage in ELEKT-D and the no-GA-required nature of ketamine, not any catastrophic-risk concern about ECT.
- **If Dad actively wants to defer ECT** and try ketamine: the A2 restart on June 8 is the active path; the U-M ECT team can be a held-in-reserve resource if ketamine doesn't work.
- **What probably shouldn't drive the decision:** family preference, the desire to "do something" after the difficult 5/18 session, or unstated assumptions about which treatment is the "real" intervention. Both ECT and ketamine are real interventions with real evidence; the question is which fits Dad's preferences and situation best.

A CONSIDERATION THAT MAY MATTER REGARDLESS OF WHICH PATH IS CHOSEN

Completing the DAT-SPECT and the U-M Movement Disorders evaluation in the next few weeks would meaningfully refine the picture either way: if Dad ends up doing ECT, abnormal DAT-SPECT confirms him in the Morikawa cohort (favorable outcomes) and informs technique; if Dad ends up doing ketamine, the same workup informs the depakote question and the longer-term treatment trajectory. The workup is not an obstacle to either treatment path — it's information that improves whichever path is chosen. Whether ECT can wait the 1–3 weeks for the DAT-SPECT result is a reasonable thing to ask the U-M team.

Key studies + post-consultation follow-up

ECT IN LEWY-SPECTRUM AND PARKINSON'S DISEASE — FAVORABLE SIGNAL

Morikawa F, Kobayashi R, Murayama T et al. Evaluating ECT for Dementia With Lewy Bodies, Including the Prodromal Stage.

Int J Geriatr Psychiatry 2024;39:e70020.

KEY FINDING Retrospective ECT analysis in 25 DLB / prodromal DLB / MCI-LB patients (all with abnormal DAT-SPECT and MIBG). 96% at least minimal improvement; 60% remarkable improvement. Parkinsonism improved with ECT (mean Hoehn-Yahr 1.8). Adverse events: 24% transient delirium, 4% transient amnesia; no serious AEs. **No cognitive worsening** before vs after ECT. Most directly applicable safety data for Dad's profile.

Takamiya A et al. ECT for Parkinson's Disease: Systematic Review and Meta-Analysis.

Mov Disord 2021;36:50–58.

KEY FINDING 14 studies, n=129 PD patients. ECT significantly improved both motor and non-motor symptoms without worsening cognitive functioning. No documented cases of ECT precipitating motor decline.

Kellner CH et al. Right Unilateral Ultrabrief Pulse ECT in Geriatric Depression (PRIDE Phase 1).

Am J Psychiatry 2016;173:1101–1109.

KEY FINDING n=240, age ≥60. RUL-UBP ECT plus venlafaxine: 61.7% remission in geriatric depression. Establishes ECT's real efficacy in late-life TRD with the technique that minimizes cognitive side effects.

KETAMINE COMPARISON + COGNITIVE SAFETY

Anand A et al. Ketamine vs ECT for Nonpsychotic Treatment-Resistant Major Depression (ELEKT-D).

N Engl J Med 2023;388:2315–2325.

KEY FINDING n=403. Ketamine non-inferior to ECT on response (55.4% vs 41.2%). Delayed-recall T-score change: –0.9 (ketamine) vs –9.7 (ECT) at end of treatment — meaningful cognitive-safety advantage for ketamine during the treatment course.

Kumpf KT et al. Cognitive Effects of Repeated IV Ketamine and ECT in TRD (ELEKT-D Secondary Analysis).

J Clin Psychiatry 2025.

KEY FINDING ECT cognitive cost largely transient — differences between groups resolved among responders at 1-, 3-, and 6-month follow-up. The cognitive concern is during the treatment course, not durable in responders.

GENERAL ANESTHESIA EXPOSURE CONSIDERATIONS

Schulte PJ et al. Anaesthesia, Surgery, and Long-Term Cognitive Trajectories (Mayo Clinic Study of Aging).

Br J Anaesth 2018;121:398–405.

KEY FINDING n=1,819 adults ≥50. Single new GA+surgery exposure → 0.2 SD additional cognitive decline over 5 years. Real but modest signal; applies most strongly to longer surgical GA, less cleanly to brief ECT-anesthesia.

Kunicki ZJ et al. Six-Year Cognitive Trajectory After Major Surgery and Delirium (SAGES).

JAMA Intern Med 2023;183:442–450.

KEY FINDING Postoperative delirium associated with 40% accelerated cognitive decline slope. "Delirium signals underlying brain vulnerability." The framework is conceptually important even if Dad's March course was not formally delirium.

AFTER THE U-M CONSULTATION — WHAT TO CAPTURE IN WRITING

1. What did the U-M team recommend, and what was their reasoning?
2. If ECT was recommended, what's the planned electrode placement and pulse width, anesthesia agent, and schedule?
3. What's the pre-ECT cognitive baseline plan and the serial monitoring schedule?
4. What's the team's position on waiting for DAT-SPECT results vs proceeding now?
5. What would constitute a reason to stop ECT mid-course?
6. What's the maintenance plan if Dad responds, and how does the team think about cumulative GA exposure over time?
7. How does the team think about coordinating with Dr. R, Dr. Chilton, Julane, and the U-M Movement Disorders team going forward?

A NOTE ON THE FAMILY ROLE

The family's role in this decision is to make sure Dad has the relevant information and that the U-M team has the relevant context — not to push toward a particular outcome. ECT is a reasonable treatment for Dad's severity and profile. Another ketamine course is also a reasonable treatment. The right answer depends on Dad's preferences, his appetite for each modality given recent experiences, the team's clinical judgment, and the eventual results of the workup. The family's job is to support whichever path Dad and the team choose — and to ensure the chosen path is executed with appropriate technique, monitoring, and follow-through.